

Regional climate shocks, adaptation policy and distributional outcomes in an agent-based model

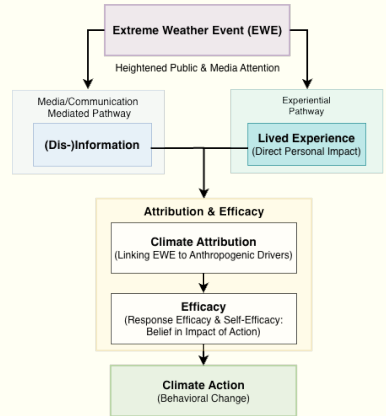
A regional version of DSK Macro-economic Agent-Based Model

Anmol Soni¹

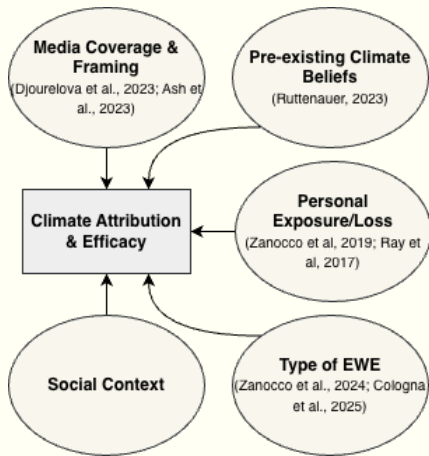
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Introduction

- ❖ Extreme weather events (EWEs) are increasing in frequency, intensity and economic damages
Intergovernmental Panel On Climate Change (2014); Coronese et al. (2019)
- ❖ Public and media attention are heightened during EWE
Roxburgh et al. (2019); Friedman et al. (2019)
- ❖ EWEs can serve as catalysts for major societal and political shifts (“critical junction”) Casali et al. (2024)
- ❖ Attribution of EWEs to climate change is key for converting experience into mitigation motivation
Ogunbode et al. (2019)



What shapes climate attribution post EWE?



Heterogeneity in climate impacts & opinions

- ❖ Climate damages from EWEs are **uneven across regions**, creating long-run path dependencies in economic performance and climate policy support

Ray et al. (2017); Giordono et al. (2020a)

- ❖ Regional EWE exposure differs in frequency as well as type
- ❖ National support for climate policy reflects interactions across regions, including:
 - ❖ **Opinion dynamics** among affected residents (e.g., climate attribution after shocks)
Cologna et al. (2025)
 - ❖ **Economic & demographic weight** of impacted regions (core vs periphery)
Bølstad (2015)
- ❖ Heterogeneous exposure & perceptions help explain **variation in local support for climate policies**

Giordono et al. (2020b)

Research Question

How do regionally heterogeneous extreme weather events shape macroeconomic dynamics and climate policy outcomes through opinion dynamics?

1. How does heterogeneous exposure to extreme weather events affect regional inequalities and macroeconomic outcomes?
▶ Phase 1
2. How do localized experiences of extreme weather events influence regional and national climate policy preferences, including both the strengthening and erosion of support for mitigation and adaptation policies?
▶ Phase 2

Model Extension

Phase 1: Regionalised Agents and Shocks

- ❖ **Objective:** Explore how regional EWE exposures affect national and regional macro-economic outcomes.
- ❖ **Modelling:**
 - ❖ Regionalised agents: firms, labour (households), and energy plants
 - ❖ Capital Market, Goods Market and Energy Market remains at national level (unchanged).
 - ❖ Regional governments with budgets and policies.
- ❖ **Hypothesis:** Regional EWE damages create persistent economic path dependencies.
- ❖ **Policies:**
 - ❖ Baseline: no active regional governments.
 - ❖ Adaptation: regions offer relief subsidies or invest in damage reduction/rebuilding post EWE.
 - ❖ Policy mixes: test interactions of national mitigation and regional adaptation.

Phase 2: Regional Shocks & OD

❖ **Objective:** Endogenize support for climate policies via opinion dynamics.

❖ **Modelling:**

- ❖ Drivers: EWE exposure, attribution, beliefs, employment, (dis)information.
- ❖ Outcomes: Differentiated support for mitigation (national) vs. adaptation (regional) policies.

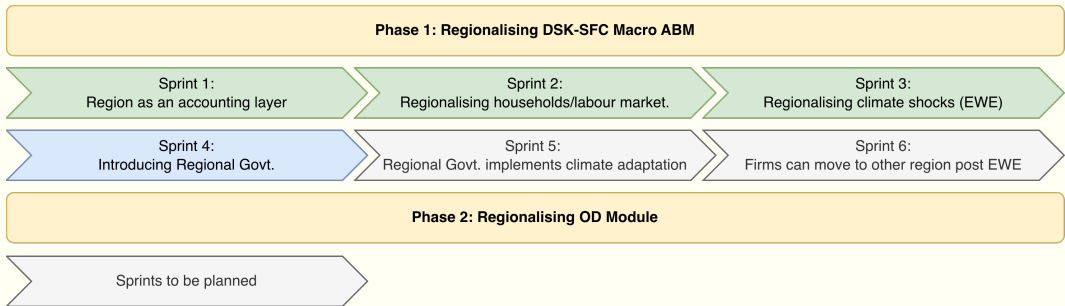
❖ **Hypotheses:**

- ❖ EWE exposure increases adaptation support; mitigation support is conditional
[▶ see figure](#)
- ❖ Repeated EWE can either exacerbate or alleviate political polarization and climate fatalism among residents.

❖ **Policies:**

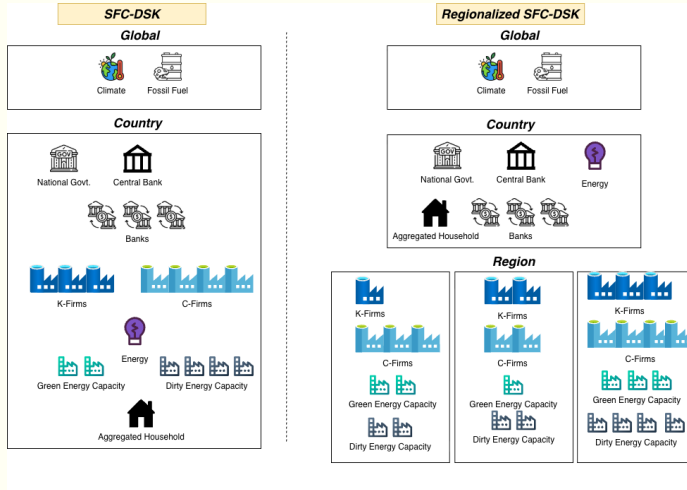
- ❖ Baseline: Households form opinions on national carbon policy.
- ❖ Adaptation scenario: Opinions include regional adaptation support.
- ❖ Election scenario: Opinion shifts only affect policies during election cycles.

Model Extension Roadmap: Sprint Planning



Regional Firms

Regionalising K & C Firms and Energy Capacity



Regional Layer

1. Region is conceptualized as an aggregator/accounting layer in the existing model. It is not an agent and does not influence model behaviour.
2. In this first sprint, K-Firms, C-Firms are tagged to a region.
3. Energy capacity, production, emissions, etc. are computed at regional level as derivatives based on regional capacity share.
4. **'AGGREGATE_REGIONS'** function is introduced which reports regional aggregates and also act as a verification to ensure regional aggregates sum to global aggregates.
5. Other variables and behavioural aspects are left untouched.

Regionalising K & C Firms

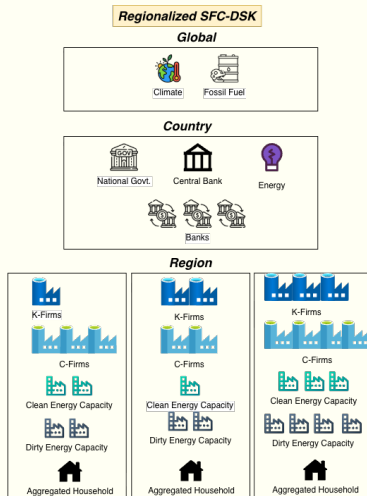
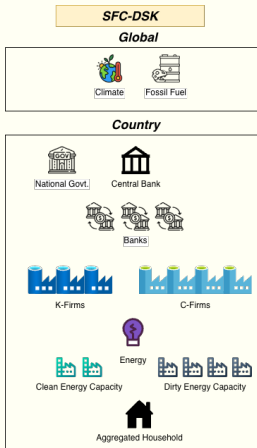
- ❖ If “**regions**” entry is present in the input file, firms are assigned a regional tag at initialization (sequential attribution).
- ❖ Upon exit and entry, firms inherit the regional tag associated with the slot they occupy.
- ❖ **Assumption** the proportion of K & C-firms does not evolve endogenously over the time.
- ❖ **Assumption** The market remains at national level. **No** regional dynamics here!

Regionalising Energy Plants

- ❖ Regional accounting layer computes following energy sector parameters (derived) at regional level using `dirty_capacity_share` and `green_capacity_share`.
- ❖ The energy market is also at national level analogous to a national electricity grid.
- ❖ The climate shock to energy capacity (plants) can be implemented using `dirty_capacity_share` and `green_capacity_share` when a region is hit by an EWE/Shock.

Regional Labour

Regional Labour / Households



Refining regional allocation of firms & energy

- ❖ Introduced proportion-based regional allocation for:

1. K-firms (static)
2. C-firms (static)
3. Dirty energy capacity & expansion
4. Clean energy capacity & expansion

- ❖ Endogenous regional distribution of Labour ‘LS0’

- ❖ Regional computation of Emissions from K-Firms, C-Firms & Energy sector

Excerpt from ‘regions_input.json’

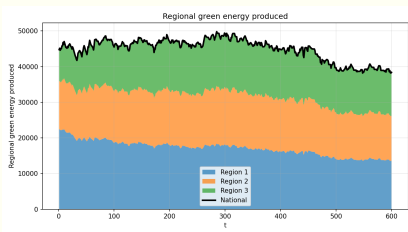
```
...
"regions":
{
  "NR": 3,
  "K_firm_shares": [0.3, 0.2, 0.5],
  "C_firm_shares": [0.3, 0.2, 0.5],

  "energy": {
    "dirty_capacity_shares": [0.1, 0.2, 0.7],
    "green_capacity_shares": [0.5, 0.3, 0.2]},

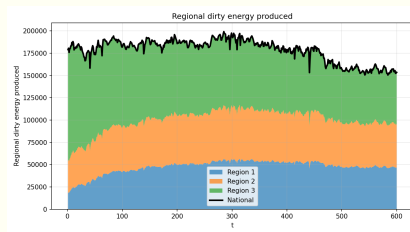
  "de_growth_probability": [0.1, 0.2, 0.7],
  "ge_growth_probability": [0.5, 0.3, 0.2]
}
...
```

Regional energy growth capacity

- ❖ Distribution of clean and dirty energy capacity is specified in the 'regions_input.json'
- ❖ New capacity expansion is determined based on regional growth probability specified in 'regions_input.json'.



Green Energy Production



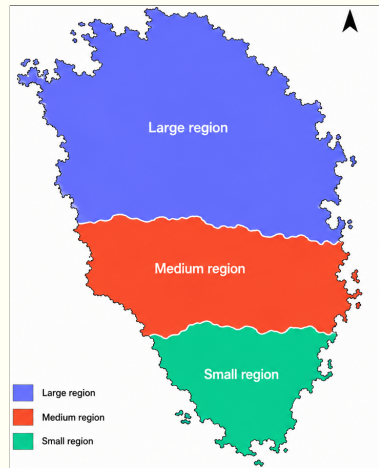
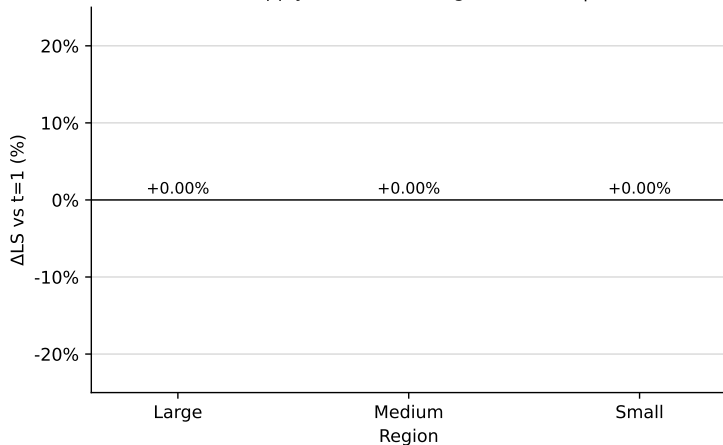
Dirty Energy Production

Regionalising Households/Labour

1. Regional labour demand (C-Firms, K-Firms and Energy Sector) is computed during initialisation and Labour 'LS0' is distributed across regions in a way that all labour demands are met at 1.
2. Regional labour shares are distributed endogenously based on labour demand by K and C firms in that region. This keeps the existing SFC-DSK mechanisms intact.
3. **Assumption** **Free inter-region mobility** If a labour loses employment, it will move to another region with labour demand.
4. Labour stays unemployed in a region, until a new labour demand emerges in the same or different region.

Inter-regional labour mobility

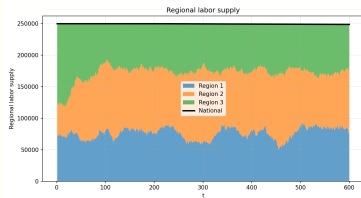
Labour Supply - relative change from $t=1$ | $t = 1$



Large : Small : Medium = 0.5 : 0.3 : 0.2

Regionalising Households/Labour

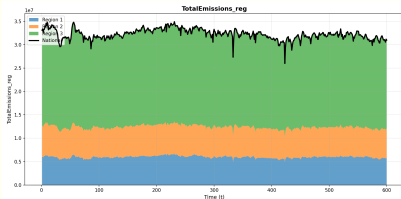
Labour Supply



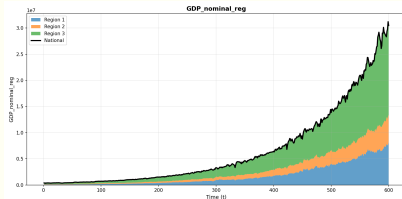
Employment Rate



Emissions



Nominal GDP



Regional Shocks

Climate Shock Channels in DSK

- ❖ All firms in the region receive uniform shocks `flag_uniform_shock = 1`
 1. Shocks to productivity affecting the characteristics of capital vintages `flag_prodshocks1`
 2. Shocks to productivity (not affecting characteristics of capital vintages) `flag_prodshocks2`
 3. Energy Capacity both green and brown `flag_encapshocks`
 4. Shock to labour population `flag_popshocks`
 5. Shock to aggregate consumption demand `flag_demandshocks`
 6. Shock to C-Firms capital stock `flag_capshocks`
 7. Output shocks to C & K Firms `flag_outputshocks`
 8. Shock to C-Firms inventories `flag_inventshocks`
 9. Shock to R & D of K Firms `flag_RDshocks`

Climate Shocks in DSK

Climate shocks are modelled as stochastic shocks drawn from temperature-dependent Beta distributions in Reissl et al. (2025).

For each shock channel $s \in \{1, \dots, S\}$ and period t :

$$\varepsilon_{s,t} \sim \text{Beta}(a_{s,t}, b_{s,t}), \quad \varepsilon_{s,t} \in (0, 1)$$

The shape parameters evolve with global temperature T_t :

$$a_{s,t} = a_{0,s} \left[1 + \ln \left(\frac{T_{t-1}}{T_0} \right) \right]^{\theta_s^{(1)}}$$

$$b_{s,t} = b_{0,s} \left(\frac{T_0}{T_{t-1}} \right)^{\theta_s^{(2)}}$$

- ❖ $a_{0,s}, b_{0,s}$ are baseline shape parameters
 - ❖ $a_{s,t}$ governs mean/median of damages
 - ❖ $b_{s,t}$ governs right-tail thickness (extreme events)
- ❖ $\theta_s^{(1)}$ controls increase in typical damages
- ❖ $\theta_s^{(2)}$ controls amplification of extremes
- ❖ T_t is global mean temperature, T_0 is reference level
- ❖ Climate shocks affect real economic variables multiplicatively:

$$X_{i,t} = X_{i,t-1} (1 - \varepsilon_{s,t})$$

Parameters of β curves for drawing shocks

1. **Global temperature** (T_t) is common to all regions
2. **Shock exponent** $\theta_s^{(1)}$ **and baseline severity parameters** $a_{0,s}$ have regional values but are kept same common across regions as they reflect physical damage sensitivities (how temperature translates into damage)
3. **Baseline exposure** ($b_{0,r}$): Region-specific right-tail thickness capturing structural exposure to extreme events (e.g. coastal vs inland regions).
4. **Extreme-event amplification** ($\theta_r^{(2)}$): Region-specific scaling of tail risk as temperature rises, allowing extreme damages to accelerate faster in some regions.

Regionalisation of Climate Shock Channels

- ❖ In the regional extension, climate shocks are generated *at the regional level* and propagated to all firms located in that region.
- ❖ For each shock channel $s \in \{1, \dots, S\}$, region r , and period t :

$$\varepsilon_{s,r,t} \sim \text{Beta}(a_{s,r,t}, b_{s,r,t})$$

- ❖ The realised regional shock $\varepsilon_{s,r,t}$ is then broadcast to all firms (or specified channel) i in region r :

$$X_{i,t} = X_{i,t-1}(1 - \varepsilon_{s,r,t}), \quad i \in r$$

- ❖ This mechanism is identical to Reissl et al. (2025).

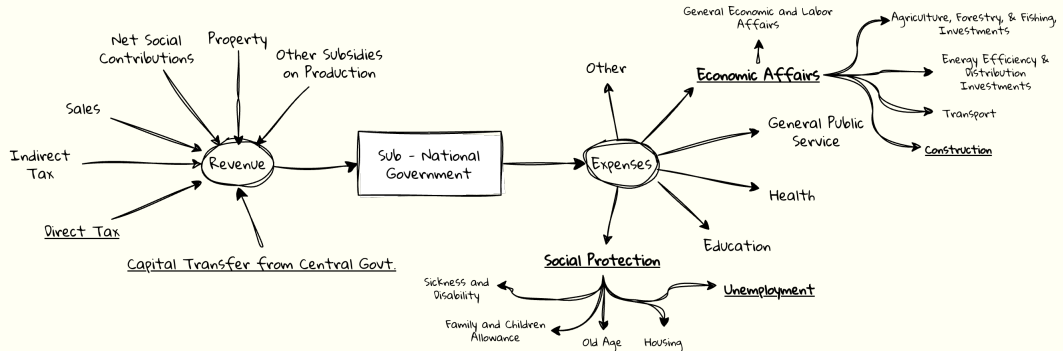
Regionalisation of climate shock channels

- ❖ At each period, we draw one climate shock per region and per channel, driven by a common global temperature path and applied uniformly to all firms in the region.
- ❖ Regions differ in how strongly specific climate shock types hit (damages) and scale with warming, but all regions share the same global climate forcing.

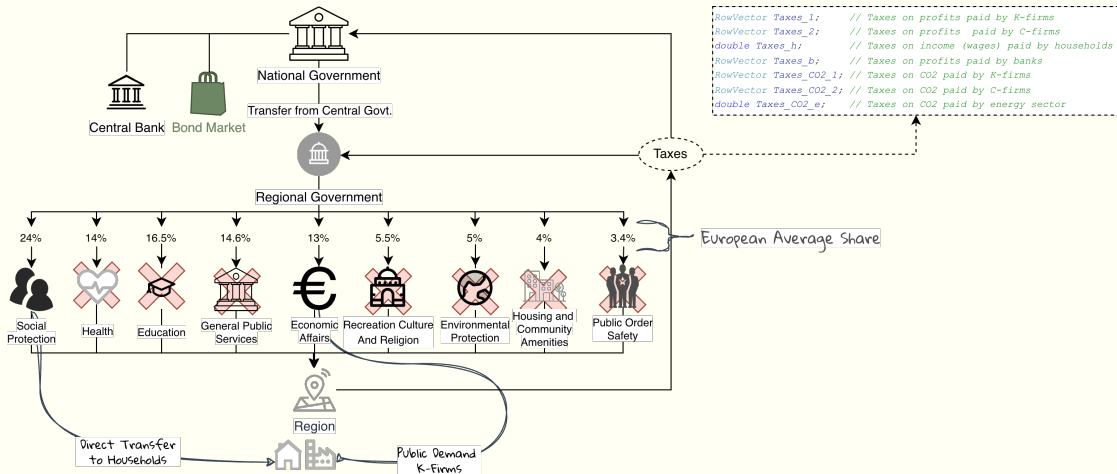
Regional Government

Conceptualising Sub-National Government

- ❖ Each country creates its own arrangement for dividing responsibilities between regional and local entities which can be collectively called as sub-national governments (Halaskova et al., 2021)
- ❖ Expenses and revenues of sub-national government can be calibrated using subnational government finances from ECB Olsson and Catz (2023)



Regional Govt. Revenue & Expenditure



Regional Government Revenue Design

$$TaxShare_{r,t} = \tau_r^{share} \left(T_{r,t}^{Cfirms} + T_{r,t}^{Kfirms} \right)$$

$$GRANTPOOL_t = \bar{\gamma} \cdot REVENUE_{CentralGovt,t}$$

$$GovtTransfer_{r,t}^{base} = \omega_r \cdot GRANTPOOL_t$$

$$GovtTransfer_{r,t}^{topup} = \max \left\{ 0, SocialProtection_{r,t} - \left(TaxShare_{r,t} + GovtTransfer_{r,t}^{base} \right) \right\}$$

$$REVENUE_{RegGovt,r,t} = GovtTransfer_{r,t}^{base} + GovtTransfer_{r,t}^{topup} + TaxShare_{r,t}$$

- ❑ $TaxShare_{r,t}$: Region's share of centrally collected taxes
- ❑ $GovtTransfer_{r,t}^{base}$: Standard intergovernmental transfer from grant pool
- ❑ $GovtTransfer_{r,t}^{topup}$: Additional transfer ensuring minimum welfare provision
- ❑ $REVENUE_{RegGovt,r,t}$: Total regional government revenue
- ❑ National government collects all taxes and redistributes to regions
- ❑ Grant pool bounded: $\bar{\gamma} \in [0, 1]$
- ❑ Allocation weights: $\sum_r \omega_r = 1$
- ❑ Welfare guarantee: $REV_{rg,r,t} \geq SP_{r,t}^{plan}$

Regional Government Expenditure Design

$$EXPENDITURE_{RegGovt,r,t} = SocialProtection_{r,t} + EconomicAffairs_{r,t}$$

Social Protection

$$SocialProtection_{r,t} = LS_{r,t} \cdot U_{r,t} \cdot w_{t-1} \cdot \omega_{u,r}$$

- ❑ $LS_{r,t}$: Regional labour supply pool
- ❑ $U_{r,t}$: Regional unemployment rate
- ❑ w_{t-1} : Wages
- ❑ $\omega_{u,r}$: Regional wage replacement rate

Economic Affairs

$$EconomicAffairs_{r,t} = \max \{0, REVENUE_{r,t} - SocialProtection_{r,t}\}$$

- ❑ Public investment expenditure for building public stock
- ❑ For each K-firm in region r :

$$EconomicAffairs_{Kfirm,r,t} = \frac{EconomicAffairs_{r,t}}{N_{r,t}^{Kfirm}}$$

Adaptation

Conceptualising Climate Adaptation in R-DSK

- ❖ DSK (Reissl et al., 2025) has climate shock propagation and mitigation policy levers but no explicit climate adaptation

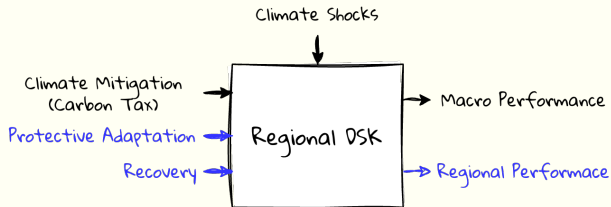
❖ Objective

- ❖ Integrate explicit, regional, and dynamic adaptation
- ❖ Preserve stochastic micro-damage structural approach of climate shock function
- ❖ Ensure stock-flow consistency

❖ Approach

- ❖ Adaptation modifies:

- ▶ Realised damage (through protection)
- ▶ Recovery dynamics (re-building)



Climate Adaptation: Conceptual Approaches

1. Adaptation as residual (implicit)

- Embedded in aggregate damage functions (net damages)
- No explicit modelling of behaviour, policy, or capacity; cannot analyse adaptation decisions or heterogeneity Patt et al. (2010)

2. Adaptation as an optimisation decision

- Trade-off between adaptation investment and residual damages
- Includes *flow* (reactive) and *stock* (capacity-building) adaptation, but remains highly aggregated and representative-agent based (Bruin et al., 2007)

3. Adaptation as sectoral and spatially heterogeneous

- Implemented in specific sectors or regions (e.g. agriculture, coastal, urban systems)
- Recognises the local and context-specific nature of adaptation processes Füssel (2010); Meng et al. (2020)

4. Adaptation as dynamic capacity and coping range

- Damages depend on climate relative to a time-varying coping range
- Distinguishes *exposure* from *adaptive capacity*; captures acclimatisation and resilience dynamics (Estrada et al., 2019)

5. Adaptation as constrained by capacity, institutions, and barriers

- Adaptation depends on finance, governance, knowledge, and implementation capacity
- Barriers prevent adaptation even when it is welfare-improving Eisenack et al. (2014); Andrijevic et al. (2023)

6. Adaptation as limited and potentially maladaptive

- Adaptation can be path-dependent, uneven, and subject to hard or soft limits
- Some responses may reduce risk in the short run while increasing long-run vulnerability Berkhout and Dow (2023); Reckien et al. (2023)

Regional Adaptation Spending Rule

- ❖ Adaptation is modelled as a combination of planned capacity-building (protective) and recovery (reactive) expenditures, where investment in adaptation can contribute to the accumulation of resilience over time (Bruin et al., 2007; Füssel, 2010; Monsod et al., 2023).
- ❖ Public spending increases after disasters (Hochrainer-Stigler et al., 2020; Bayar and Yarbrough, 2024).
- ❖ Adaptation capacity is strongly linked to income and fiscal space (Saeed et al., 2023).

Regional Government Adaptation Spending

Regional Government Expenditure

$$EXP_{r,t} = SP_{r,t} + I_{r,t}^{\text{Adapt}} + EA_{r,t}^{\text{res}}$$
$$EA_{\text{RegGovt},r,t}^{\text{res}} = \max \left\{ 0, REV_{r,t} - SP_{r,t} - I_{r,t}^{\text{Adapt}} \right\}$$

$EA_{r,t}^{\text{res}}$ is residual productive expenditure after adaptation commitments are met.

Adaptation investment

$$I_{r,t}^{\text{Adapt}} = I_{r,t}^{\text{Prot}} + I_{r,t}^{\text{Rec}}$$
$$I_{r,t}^{\text{Prot}} = \iota_r^A \cdot Y_{r,t-1}$$
$$I_{r,t}^{\text{Rec}} = \psi_r^A \cdot Y_{r,t-1} \cdot \max \left(0, \tilde{S}_{r,t-1} - \bar{s}_r \right)$$

Central Government Funding Ex-Post Recovery

If fiscal space is insufficient to fund $I_{r,t}^{\text{Rec}}$, the central government extends a recovery top-up from the national grant pool ($GRANTPOOL_t = \bar{\gamma} T_t$):

$$GT_{r,t}^{\text{Rec}} = \max(0, I_{r,t}^{\text{Rec}} - \max(0, REV_{r,t} - SP_{r,t}))$$

This prevents the perverse feedback in which large shocks simultaneously destroy regional tax revenues *and* make recovery spending infeasible.

Grant pool priority ordering

$$\underbrace{GT_{r,t}^{\text{SP}}}_{\text{SP top-up (existing)}} \succ \underbrace{GT_{r,t}^{\text{Rec}}}_{\text{Recovery top-up (new)}} \succ \underbrace{GT_{r,t}^{\text{base}}}_{\text{Base grant for Protective Adaptation / EA}}$$

Both SP and recovery are guaranteed obligations of the centre; planned protection is funded from the residual base allocation.

Protection: Ex-Ante / Planned Adaptation

$$I_{r,t}^{Prot} = \iota_r^A \cdot Y_{r,t-1}$$

- ❖ ι_r^A : exogenous region-specific planned protection intensity (share of lagged real output); scaled so that larger/wealthier regions sustain proportionally larger protection budgets
- ❖ Implemented as a government order placed with K-firms in the region: the government credits K-firm deposit accounts by $I_{r,t}^{Prot} / N_{1,r,t}$ per firm, generating demand for physical protection capital (flood defences, drainage, resilient infrastructure) in the capital-goods sector

Protection (adaptation) capital accumulation

$$K_{r,t}^{Prot} = (1 - \delta_{Prot}) K_{r,t-1}^{Prot} \cdot I_{r,t}^{Prot}$$

Protection: Climate Shocks under Adaptation

Exposure multiplier

$$\Omega_{r,t} = \bar{\omega}_r + (1 - \bar{\omega}_r) \exp \left(-\phi_P \frac{K_{r,t}^{\text{Prot}}}{Y_{r,t-1}} \right) \in (\bar{\omega}_r, 1]$$

- ❖ $\Omega_{r,t} = 1$: no accumulated protection ($K_{r,t}^{\text{Prot}} = 0$) — full exposure
- ❖ $\Omega_{r,t} \rightarrow \bar{\omega}_r > 0$: maximum protection stock — **residual exposure always remains**; even perfect infrastructure does not eliminate stochastic losses
- ❖ $\phi_P > 0$: efficiency of protection capital in reducing exposure (calibrated parameter)

Effect on realised shocks

$$shock_{r,j,t}^{\text{eff}} = \Omega_{r,t} \cdot shock_{r,j,t}^{\text{raw}}$$

$$shock_{r,j,t}^{\text{raw}} \sim \text{Beta} (Xa_{r,t}^s, Xb_{r,t}^s)$$

Protection compresses the shock realisation by $\Omega_{r,t} < 1$, but since $shock_{r,j,t}^{\text{raw}}$ is drawn from a Beta distribution, large realisations still pass through. Thus, protection is probabilistic, not deterministic.

Regions and Scenarios in the Experiment

❖ Regions:

- ❖ We create three heterogeneous regions $R1 : R2 : R3 = 0.3 : 0.2 : 0.5$.
- ❖ K-Firms, C-Firms and Energy assets are distributed across regions as per aforementioned ratio.
- ❖ We test “Smaller Region, More Exposed” i.e. $R2$ face higher climate shocks (fatter tail).

❖ Scenarios Compared:

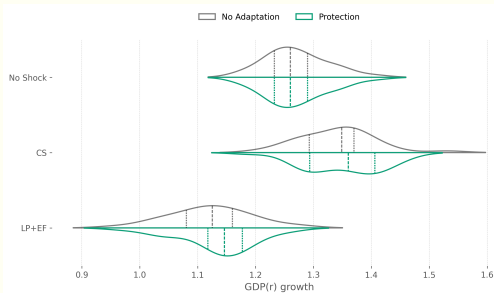
❖ No Adaptation:(DSK Baseline)

- ▶ Regions do not invest in protection.
- ▶ All regions are fully exposed to climate shocks.

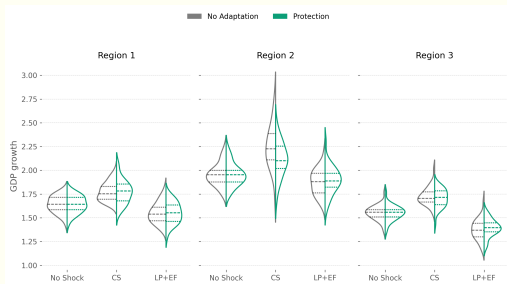
❖ Protection (Adaptation):

- ▶ Regions invest in protective infrastructure (e.g., flood defences).
- ▶ Smaller region (more exposed) spends more
- ▶ Adaptation spending as percentage of regional GDP for $R1$, $R2$ and $R3$ resp. 0.5%, 1%, 0.5%
- ▶ This investment reduces the impact of climate shocks.

Protective Adaptation: GDP Growth



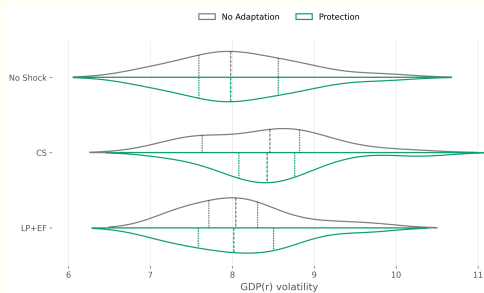
Macro



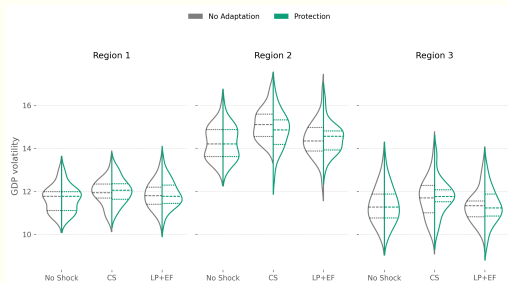
Regional

The results represent 30 Monte Carlo simulations over 600 time steps. The figures shown above correspond to the interval $200 < t < 600$.

Protective Adaptation: GDP Volatility



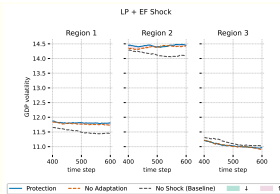
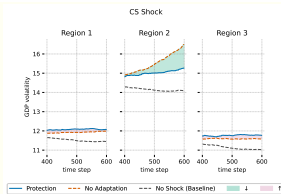
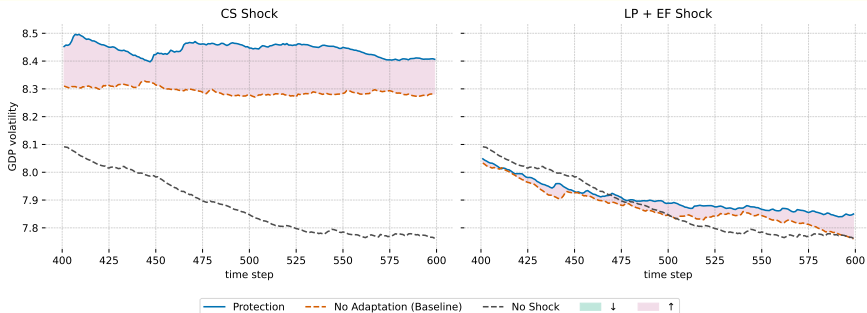
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Regional

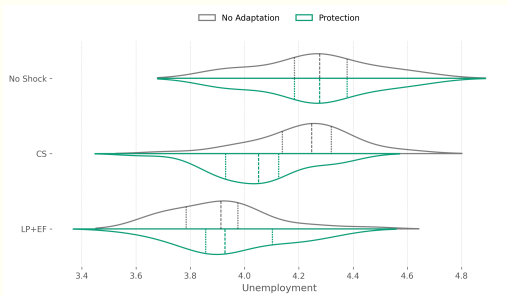
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Protective Adaptation: GDP Volatility (trend)

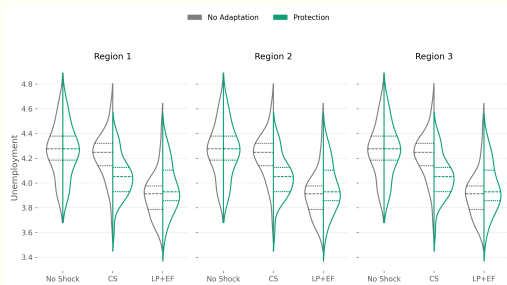


The results represent 30 Monte Carlo simulations over 600 time steps. The figures shown above correspond to the interval $400 < t < 600$.

Protective Adaptation: Unemployment



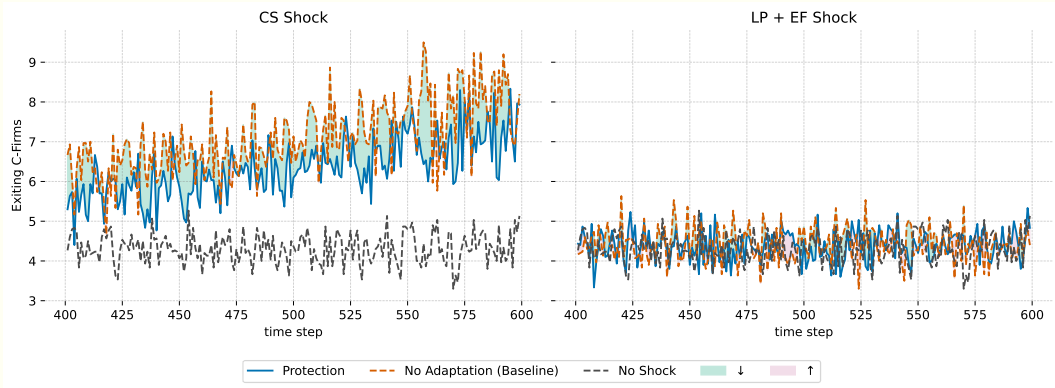
Macro



Regional

The results represent 30 Monte Carlo simulations over 600 time steps. The figures shown above correspond to the interval $200 < t < 600$.

Protective Adaptation: Exiting C firms (trend)



The results represent 30 Monte Carlo simulations over 600 time steps. The figures shown above correspond to the interval $400 < t < 600$.

Recovery Expenditure

- ❖ Capital shock (e.g. floods) destroys capital but no *recovery* mechanism via govt. intervention
- ❖ However, post-flood public expenditure rises for rebuilding and recovery.
- ❖ E.g. Italy 2023: **€4.7bn** mobilised to cover for **€10bn+** damage $\Rightarrow$ coverage ratio $\approx$ **47%** (Lodi et al., 2023)
 - ❖ 17–23% for at 6–18 months
 - ❖ Financed almost entirely via intergovernmental transfers & EU Solidarity Funds

Key modelling implication

Recovery is conceptualised as a **direct monetary transfer** to affected C-firms' deposit accounts, financed by transfers from higher government levels.

Recovery Transfer to Affected C-Firms

Revised per-firm subsidy

Define the affected firm set as those with a realised capital shock above a minimum firm-level damage share $\bar{d}$:

$$\mathcal{J}_{r,t-1}^{\text{aff}} = \{j \in r : s_{j,t-1}^{\text{cap,eff}} \geq \bar{d}\}$$

The per-firm subsidy is then:

$$\text{sub}_{j,t}^{\text{Rec}} = \frac{G_{r,t}^{\text{RecPaid}}}{|\mathcal{J}_{r,t-1}^{\text{aff}}|} \cdot \mathbf{1}(j \in \mathcal{J}_{r,t-1}^{\text{aff}})$$

Recovery backlog state variable

At each period t , a new recovery obligation $I_{r,t}^{\text{Rec}}$ is added to backlog $B_{r,t}$, which disburses at rate δ_r^{imp} :

$$B_{r,t} = I_{r,t}^{\text{Rec}} + (1 - \delta_r^{\text{imp}}) B_{r,t-1}$$

$$G_{r,t}^{\text{RecPaid}} = \delta_r^{\text{imp}} B_{r,t}$$

Recovery Expenditure Equations

Step 1 | Recovery obligation

$$I_{r,t}^{\text{Rec}} = \psi_r^A \gamma_{r,t-1} \cdot \max(0, S_{r,t-1}^{\text{aff}} - \bar{s}_r) \cdot \mathbf{1}[\psi_r^A > 0]$$

Step 3 | Per-firm subsidy (affected firms)

$$\text{sub}_{j,t}^{\text{Rec}} = \frac{G_{r,t}^{\text{RecPaid}}}{|\mathcal{J}_{r,t-1}^{\text{aff}}|} \cdot \mathbf{1}[j \in \mathcal{J}_{r,t-1}^{\text{aff}}]$$

Step 2 | Backlog update and disbursement

$$B_{r,t} = I_{r,t}^{\text{Rec}} + (1 - \delta_r^{\text{imp}}) B_{r,t-1}$$
$$G_{r,t}^{\text{RecPaid}} = \delta_r^{\text{imp}} B_{r,t}$$

Step 4 | Central government backstop

$$T_{r,t}^{\text{REC}} = \max(0, G_{r,t}^{\text{RecPaid}} - \max(0, \gamma_{r,t}^{\text{tax}} - G_{r,t}^{\text{SP}}))$$

Step 5 | Residual to public investment and adaptation

$$R_{r,t} = \max(0, \gamma_{r,t}^{\text{tax}} + T_{r,t}^{\text{CG}} - G_{r,t}^{\text{SP}} - G_{r,t}^{\text{RecPaid}})$$

$$G_{r,t}^{\text{PA}} = \min(R_{r,t}, \bar{G}_{r,t}^{\text{PA}}) \cdot \mathbf{1}_{\text{PA}}$$

$$G_{r,t}^{\text{EA}} = \max(0, R_{r,t} - G_{r,t}^{\text{PA}})$$

Recovery Expenditure | Transaction-Flow

Flow	Central Govt	Regional Govt	C-Firms (affected)	Banks	Row sum
Recovery transfer $TREC_{r,t}$	$-TREC$	$+TREC$			0
RG recovery disbursement $GRecPaid_{r,t}$		$-GRecPaid$	$+GRecPaid$		0
CG bond issuance (financing)	$+TREC$			$-TREC$ (bond held)	0
Net position	0	$TREC - GRecPaid$	$+GRecPaid$	$-TREC$	0

Bank reserve inflows ($+GRecPaid$) created by the CB during settlement are omitted for brevity; the CB absorbs the matching liability.

Recovery Expenditure

C-firm direct grant

Affected C-firms receive $subRec_{j,t}$ as a direct deposit credit to their bank account. Banks receive matching reserve inflows via CB settlement. **No K-firm machine orders are placed** as part of the recovery mechanism.

SFC check: C-firm net worth

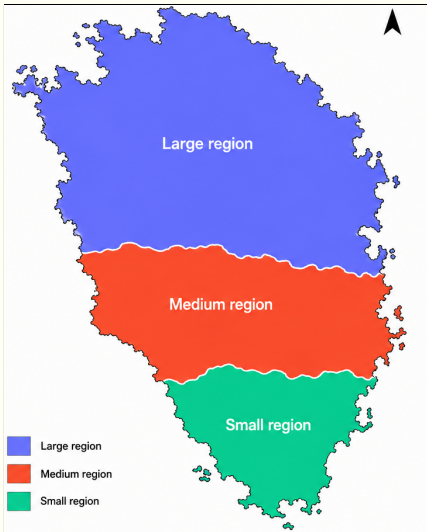
The deposit credit enters the C-firm NW flow identity:

$$\widetilde{NW}_{j,t} = NW_{j,t-1} + \Pi_{j,t} + subRec_{j,t} + \dots$$

Capital stock $K_{j,t}$ is **not** updated by the grant.

Simulation Experiment

Model Setup



Model setup

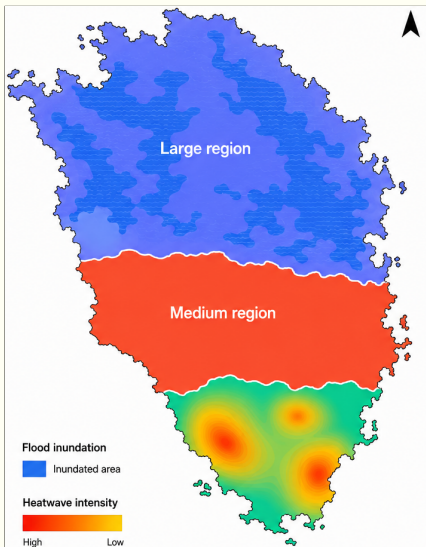
- ❖ $T = 600$ (Climate box starts at $t = 200$)
- ❖ Replications: $n = 50$ Monte-Carlo simulations
- ❖ K firms: $N_1 = 20$
- ❖ C firms: $N_2 = 200$
- ❖ Regions: $N_R = 3$

Regional Setup

- ❖ *Large : Medium : Small* = 5 : 3 : 2
- ❖ These ratios are used for allocation of C-firms, K-firms, and energy sector assets

❖ [Detailed simulation setup](#)

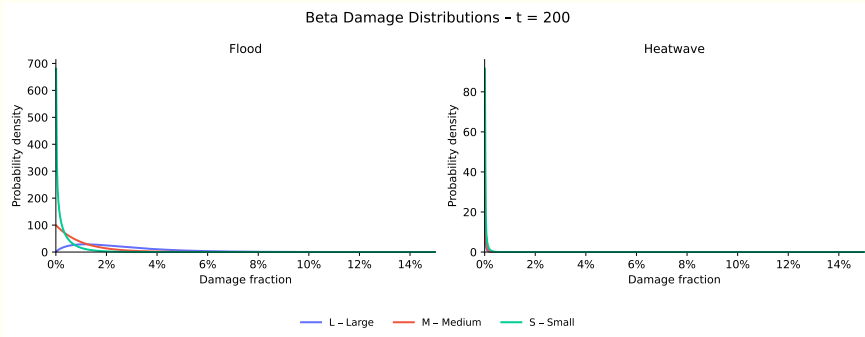
Model Setup: Climate Topology



- ❖ Large Region (North) faces $\approx 2.5\times$ higher floods exposure than Centre (Middle) and $\approx 7.5\times$ higher than Small (South) region
- ❖ Small region (South) faces $\approx 3\times$ higher heat-waves than Medium (Centre) and $\approx 7\times$ higher than large region (North)
- ❖ Shock channels
 - ❖ Floods - Capital Shock Channel
 - ❖ Heatwaves - Energy Efficiency & Labour Productivity
- ❖ Shocks are drawn every period from Beta distribution, one draw per region (uniform within region) for all active shock channels.

Model Setup: Climate Shocks

Shock parameters scale exponentially with the temperature anomaly ΔT .



Capital Shock ($\Delta T = 0$)

North	Beta(1.8, 70)	2.51%	1.83%
Centre	Beta(1.0, 100)	0.99%	0.98%
South	Beta(0.5, 150)	0.33%	0.47%

Energy Efficiency & Labour Prod. ($\Delta T = 0$)

Beta(0.003, 1350)	0.0002%	0.004%
Beta(0.005, 1000)	0.0005%	0.007%
Beta(0.01, 675)	0.0015%	0.015%

Model Setup: Simulation Scenarios

Sc.	Label	Climate Shocks		Adaptation Policy	
		Capital	Efficiency	Recovery	Protection
1	No-shock	—	—	—	—
2	Climate Shocks (Laissez-faire)	✓	✓	—	—
3	Recovery only	✓	✓	✓	—
4	Protection only	✓	✓	—	✓
5	Recovery + Protection	✓	✓	✓	✓

Model Setup: Adaptation Policies

Recovery policy (Sc. 3 & 5)

- ❖ Fiscal transfer triggered when mean regional capital loss exceeds threshold $\bar{s}_r = 2\%$
- ❖ Transfer obligation:

$$I_{r,t}^{\text{rec}} = \psi_r Y_{r,t-1} \max(0, \bar{S}_{r,t-1} - \bar{s}_r)$$

- ❖ Recovery rate: $\psi_r = 0.47$
- ❖ Backlog disbursed at $\delta_r^{\text{imp}} = 25\%$ per period
Expected implementation lag: ≈ 4 periods

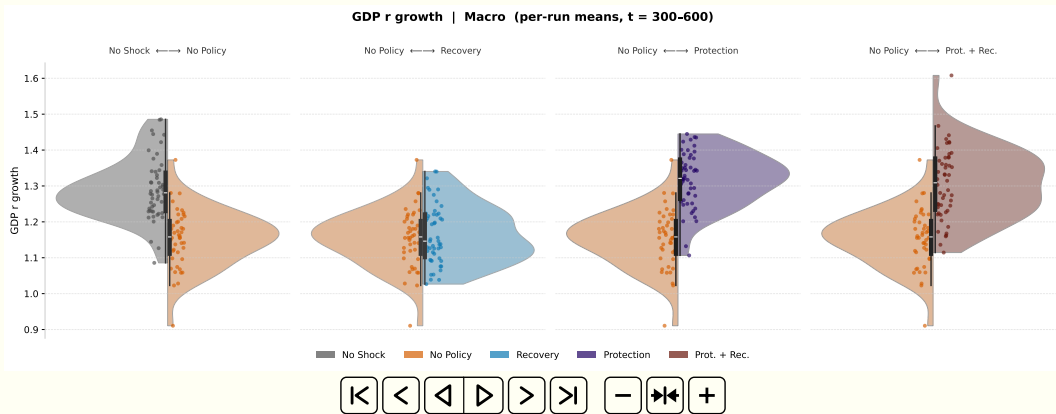
Protection policy (Sc. 4 & 5)

- ❖ Government invests $\iota_r Y_r$ in adaptation capital K_r^{ad}
- ❖ Shock dampening factor:

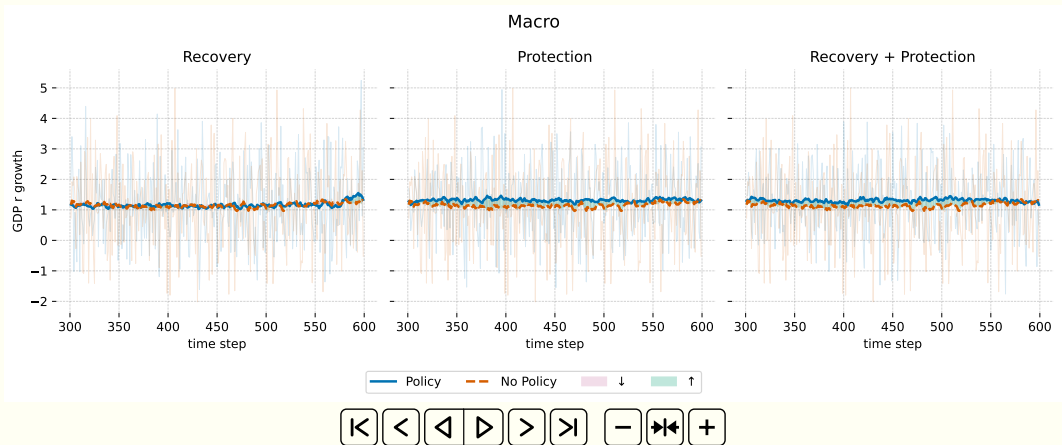
$$\Omega_{r,t} = \omega^{\text{fl}} + (1 - \omega^{\text{fl}}) e^{-\phi K_{r,t-1}^{\text{ad}} / Y_{r,t}}$$

- ❖ Investment rates: N = 0.5%, C = 1.0%, S = 0.5%
- ❖ Parameters: $\phi = 4$, $\omega^{\text{fl}} = 0.25$, $\delta^{\text{ad}} = 3\%$
Shock reduction floor: maximum dampening = 75%

GDP Growth (1/2)



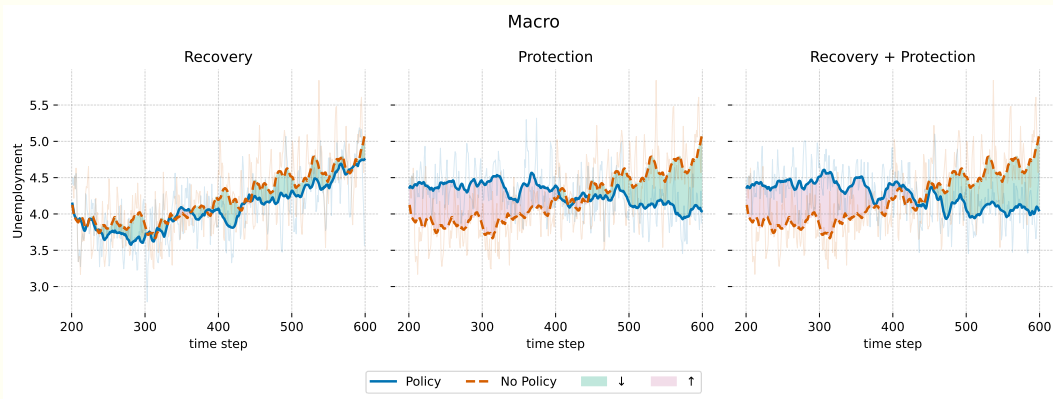
GDP Growth (2/2)



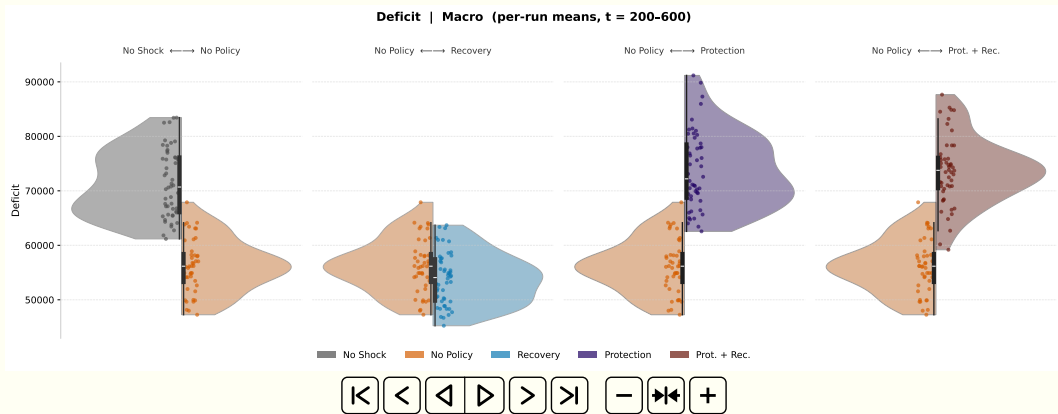
Unemployment (1/2)



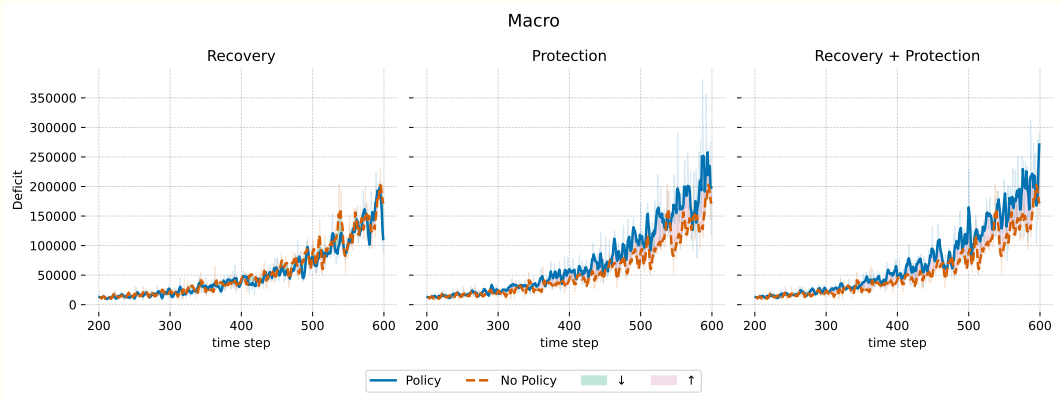
Unemployment (2/2)



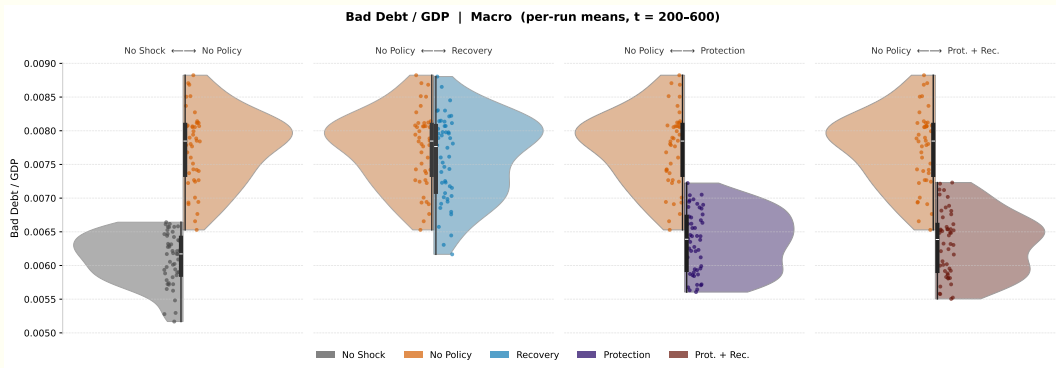
Deficit (1/2)



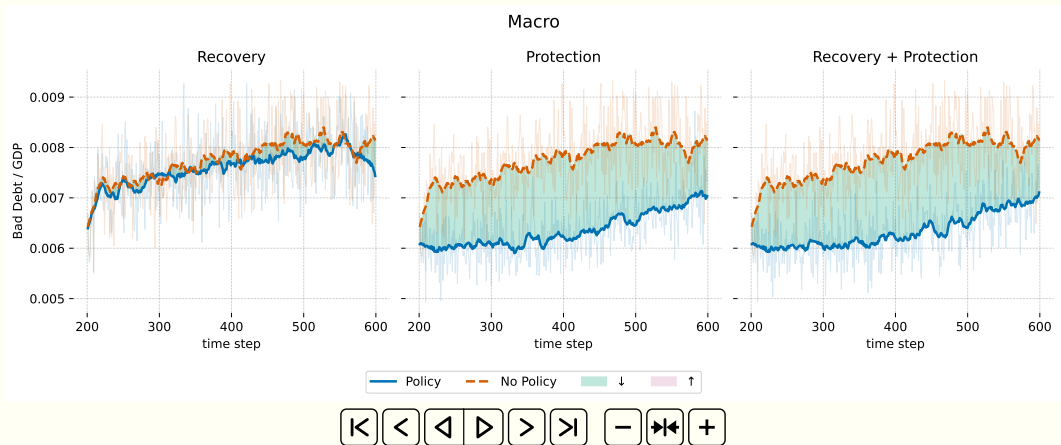
Deficit (2/2)



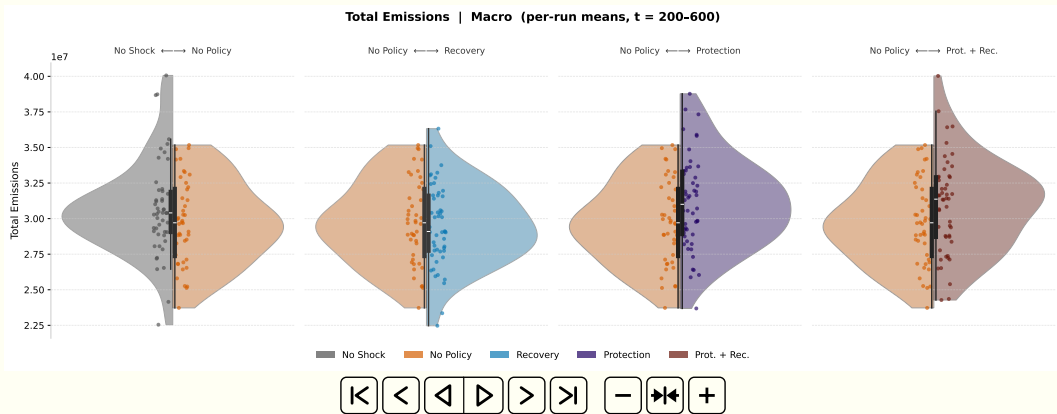
Bad Debt / GDP (1/2)



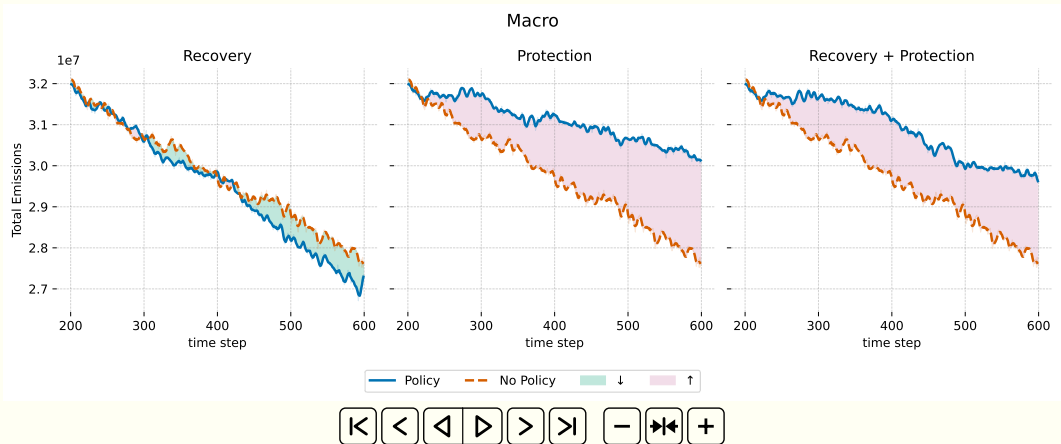
Bad Debt / GDP (2/2)



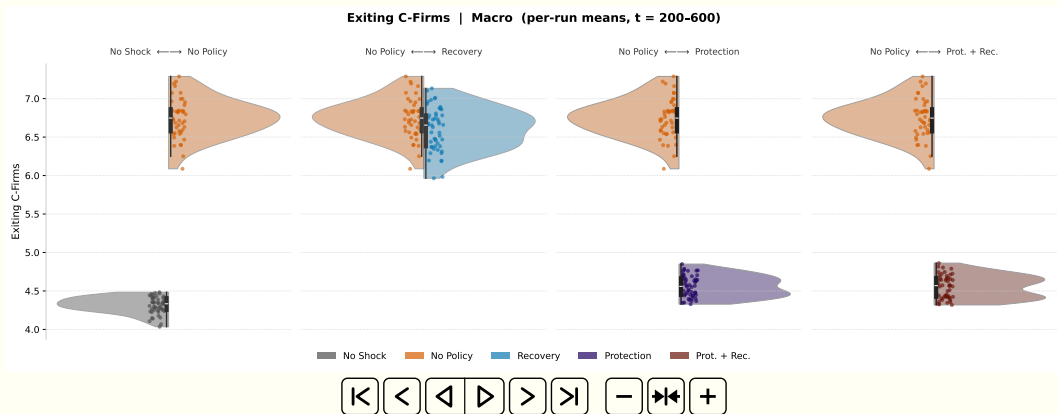
Emissions (1/2)



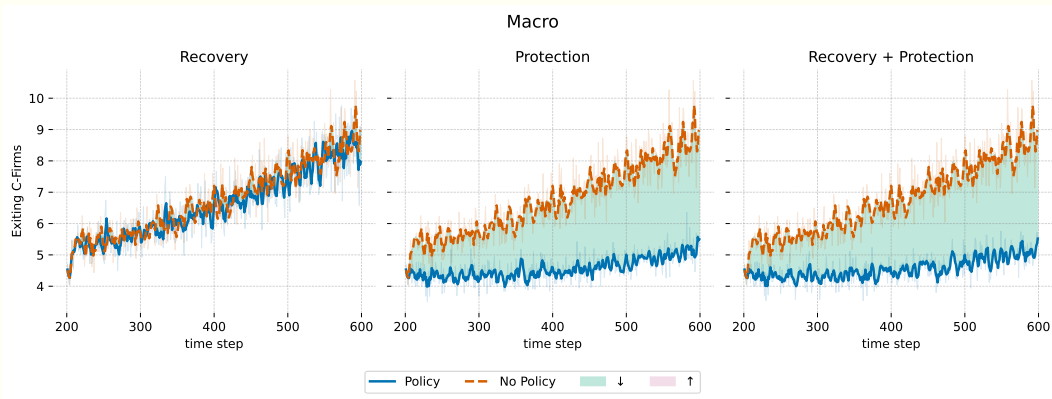
Emissions (2/2)



Exiting C-Firms (1/2)



Exiting C-Firms (2/2)



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Simulation Setup: Model Structure (1/2)

Regional Structure

- ❖ Firm shares: $[0.50, 0.30, 0.20]$
- ❖ Equal tax-revenue pooling: $\tau_r = \frac{1}{3}$
- ❖ Equal household expenditure weight: $\omega_r = \frac{1}{3}$
- ❖ Regional unemployment top-up: $\bar{w}_r^u = 0.05$

Labour Market

- ❖ Wage Phillips curve slope: $\kappa = 0.76$
- ❖ Replacement rate: $w^u = 0.35$ (35% of wage)
- ❖ Near-stationary labour supply: $g_\ell \approx 1$

Household Demand

- ❖ MPC out of income: $\alpha_1 = 0.965$
- ❖ MPC out of wealth: $\alpha_2 = 0.30$

Firms

- ❖ Capacity utilisation target: $u^* = 0.80$
- ❖ Mark-up: $\mu_1 = 0.10$ (K-firms), $\mu_2 = 0.20$ (C-firms)
- ❖ Vintage capital: maximum machine age = 19, size = 40
- ❖ Maximum investment share: $I_{\max} = 0.25$

Simulation Setup: Model Structure (2/2)

Financial Sector

- ❖ Base interest rate: $r = 4\%$
- ❖ Taylor-rule response: $\phi_\pi = 1.23, \phi_u = 0.17$
- ❖ Endogenous bank credit; no Basel floor: $\beta = 0$

Fiscal Sector

- ❖ Corporate tax rate: $\tau = 10\%$
- ❖ Public capital depreciation: $\delta_{\text{pub}} = 2.5\%$
- ❖ Regional fiscal capacity: $\bar{\gamma} = 0.20$